

Research Proposal

Association Between Diet Quality and Mental Health of Bangladeshi University Students: A Descriptive Cross-sectional Study

Background:

Depression and anxiety are widespread among young adults but somehow remain unattended for the most part of it. The relentless cycle of exams and continuous classes is bound to take a toll on mental health, leading to emotional exhaustion and burnout. Another common circumstance to see in young adults is the rapid lifestyle change leading to shift in dietary habits which are often marked by low diversity and high consumption of fast foods. These dietary patterns are associated with an increasing prevalence of non-communicable diseases like diabetes, obesity and cardiovascular diseases. Previously many studies in Bangladesh have deduced the prevalence of mental health issues among university students. According to a study by the national Institute of Mental Health in Bangladesh, around 52% of university students reported symptoms of depression, and nearly 47% experienced significant anxiety during the COVID-19 pandemic. Studies have also observed the impact of dietary diversity on overall health, where many young adults in Bangladesh consume diets dominated by rice and lack a balanced intake of essential food groups like proteins and vegetables. However, Studies linking dietary diversity with mental health specifically in young adults or university students in Bangladesh remain scarce. The objective of our study is to investigate the relationship between diet quality, non-communicable diseases (NCDs) and mental health and identify if there's any strong correlation among the three of them.

Rationale:

- Mental health disorders, including depression and anxiety, are prevalent public health issues in Bangladesh, yet remain under-researched. Studies report that the prevalence of mental disorders among adults, ranges from 6.5% to 31%, with depressive and anxiety disorders being most common.
- University students in Bangladesh show higher levels of mental health challenges compared to global averages. For example, over 50% of Bangladeshi students report moderate to extremely severe depression, while more than 58% experience similar levels of anxiety.
- Dietary diversity among Bangladeshi university students is generally low, with significant differences between public and private university students, largely due to economic disparities. Poor dietary diversity has been linked with mental health challenges in multiple studies.
- Evidence suggests an inverse association between dietary diversity and mental health issues, particularly depression and anxiety. Greater dietary diversity is associated with better mental health outcomes, as it promotes higher micronutrient intake, healthier microbiota, reduced oxidative stress, and improved immune function.
- While previous studies have explored dietary diversity and mental health links in various populations, no research has specifically examined this association in Bangladeshi university students, highlighting a gap in the literature and justifying further study in this demographic. This study aims to address this gap by investigating the relationship between diet quality and mental well-being in Bangladeshi university students.

Objectives of the Study:

Our general objective of this study focuses on these two specific questions:

- **What is the association between diet quality and mental health (in terms of depression, anxiety and stress) among university students?**
- **Does the relationship between diet quality and mental health vary across demographic characteristics?**

Research Gap:

1. **Limited Focus on Young Adults:** While there is substantial research on diet and mental health, few studies specifically focus on university students aged 20-29, who face unique stressors that may impact their mental well-being and dietary habits differently from other age groups.
2. **Insufficient Data on Diet and Mental Well-being:** Much of the current research tends to focus on clinical mental health issues (e.g., depression and anxiety) rather than overall mental well-being, which includes factors like mood, stress, and life satisfaction.
3. **Lack of Studies in Low- and Middle-Income Countries:** Most research in this area is conducted in high-income countries, and there is limited data on the relationship between diet quality and mental health among university students in low- and middle-income countries like Bangladesh.
4. **Neglect of Smoking as a Factor in Mental Well-being Studies:** Existing studies often exclude smokers, which limits understanding of how diet quality might affect mental well-being in this subgroup. By including smokers, this study addresses a gap in the current literature, examining how smoking may or may not moderate the relationship between diet and mental well-being.

Methodology:

➤ **Study Design:**

The study will be conducted through a cross-sectional study design to evaluate the association between diet quality and mental health (depression, anxiety and stress) among university students. Data will be collected at a single point from each participant by administering the survey once.

➤ **Sampling Method:**

We will use convenience sampling to recruit participants from universities of Dhaka as well as Barisal. Students will be recruited from a wide range of demographic characteristics including economic background, living arrangement (on and off-campus), academic years so as to ensure diversity

➤ **Sample size:**

A sample of 400 students will be taken to allow adequate detection of the association between mental health and diet quality.

➤ **Inclusion and Exclusion Criteria**

Inclusion Criteria:

- ✓ University students aged 20-28 years.
- ✓ Students willing to provide written consent and provide information for the survey

Exclusion Criteria:

- ✓ Students outside the specified age range
- ✓ Students who have been diagnosed with chronic depression
- ✓ Students who have had any tragic incident happen with them recently

✚ **Data Collection Tools and Assessments**

❖ **Mental Health Assessment:**

The psychological status of the participants will be measured through the **Depression, Anxiety, and Stress Scale-21** (DASS-21), a questionnaire previously validated on Bangladeshi medical university students which will assess the severity of symptoms in each of these three domains of depression, anxiety and stress.

▪ **Cutt-off values:**

	Depression	Anxiety	Stress
Normal	0-9	0-7	0-14
Mild	10-13	8-9	15-18
Moderate	14-20	10-14	19-25
Severe	21-27	15-19	26-33
Extremely Severe	28+	20+	34+

○ **Scoring:**

Scores on DASS-21 will need to be **multiplied by 2** to calculate the final score.

❖ Diet Quality Assessment:

Diet quality will be measured using the **Diet Quality-Questionnaire (DQ-Q)**. It will be assessed through:

▪ Food Group Diversity Score (FGDS):

It will measure dietary diversity by asking participants to recall their dietary intake over the past 24 hours. Each reported food group will contribute to the overall diversity score.

- Scoring: Participants will be scored from **0-10**

▪ NCD Protective and Risk Scores:

The food groups consumed over the last 24 hours will also be used to assess the NCD-protective and risk scores

- Scoring: Participants will be scored from **0-9** for both the scores respectively

▪ Global Dietary Recommendation (GDR) Score:

GDR score assesses the alignment of the dietary pattern and compares it with dietary guidelines set by the World Health Organization (WHO).

- Calculation and Scoring:

$$\text{GDR} = \text{NCD-protect score} - \text{NCD-risk score} + 9$$

It will be scores from **0-18**

▪ Demographic and Socioeconomic Data:

Data will be collected on gender, socio-economic status, employment status, academic and employment work-load, living arrangements and other potential confounders

➤ Data Collection Procedure:

Data will be collected using an online questionnaire in the form of a google form. The survey is expected to take 10-15 minutes.

➤ **Data Analysis Plan:**

❖ **Descriptive statistics:**

It will calculate the mean, standard deviations, and distributions of key variables, such as, DASS-21 scores, FGDS, NCD-risk scores, NCD-protect scores, and GDR scores.

❖ **Association Analysis:**

It will be done by using Pearson's or Spearman's correlation tests to examine the association between DASS-21 scores and each of the diet quality measure as well as other demographic variables

❖ **Group Comparisons:**

It will use t-tests or ANOVA to examine differences in the scores of the diet quality measures across different mental health levels as categorized by DASS-21

▪ **Multivariate Analysis:**

Multiple Regression model will be used to determine the association between mental health scores (DASS-21) and each measure of diet quality, while adjusting for confounding variables such as gender, living arrangements and socioeconomic status.

▪ **Software:**

Data will be analyzed using SPSS software. Statistical significance will be < 0.05

➤ **Expected result:**

- i. There is a strong correlation between dietary quality and depression, anxiety and stress.
- ii. Diet quality as well as mental health varies considerably between those with on-campus and off-campus living arrangements.

➤ **Implications:**

- i. Promotion of dietary diversity aimed to improve mental well-being of individuals.
- ii. Nutritional education from the young age and raising awareness about its impact on both physical and mental health.
- iii. Policymaking in order to ensure sufficient dietary diversity among university students living in campus hostels.
- iv. Holistic health approaches.